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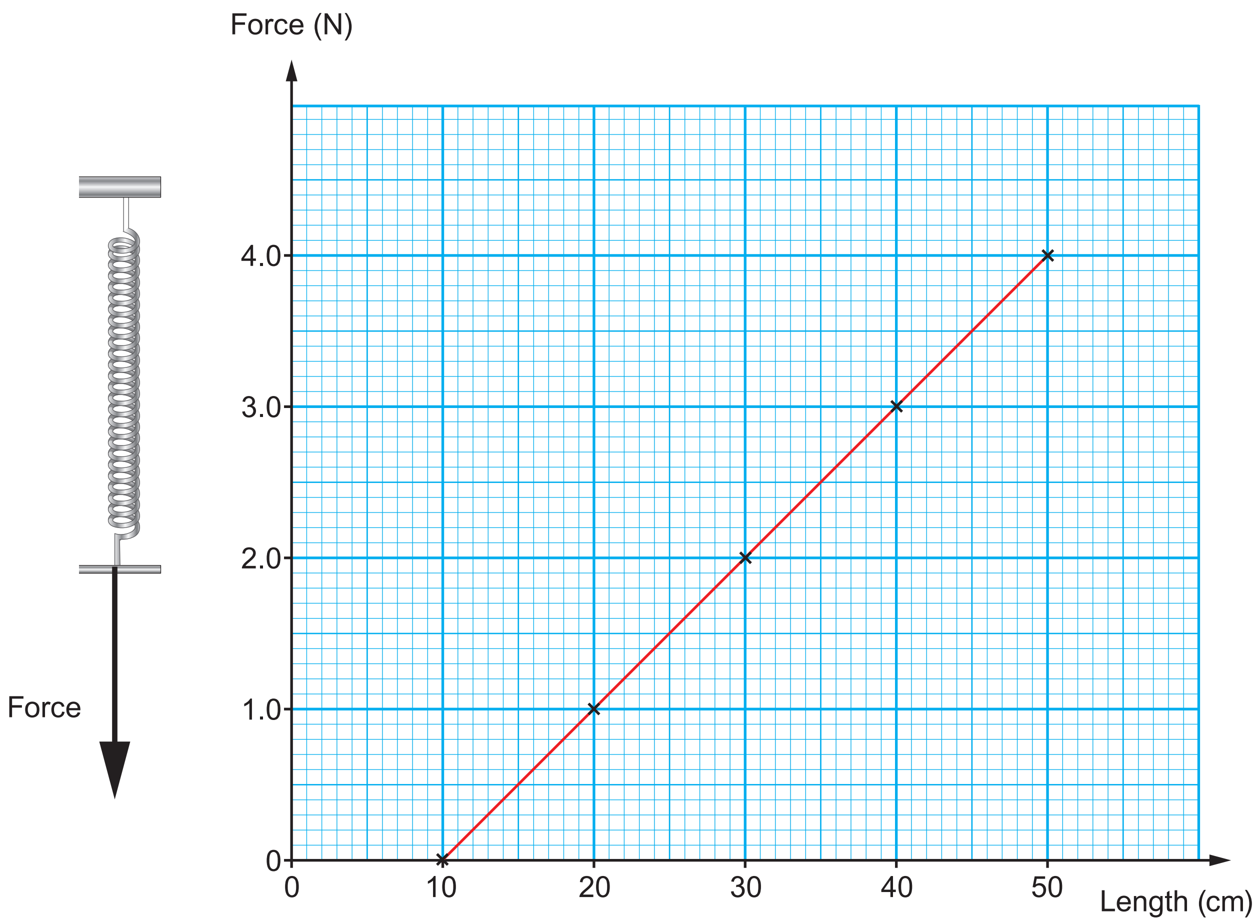
6. A group of students are investigating how a spring stretches when forces are applied to it. They measure the length of the spring and plot their results on a graph.



(a) (i) Use the graph to determine the unstretched length of the spring. [1]

Unstretched length = ... cm

(ii) Use the graph and your answer to (i) to calculate the **extension** of the spring when a force of 2.5 N is applied to it. [2]

Extension = ... cm

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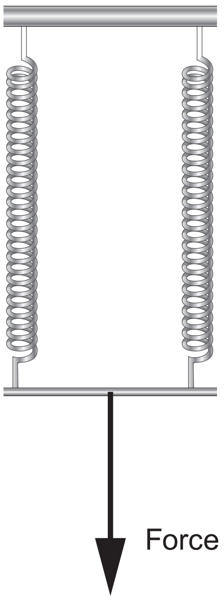
(iii) Use **your answer to (ii)** and the equation:

$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

to calculate the spring constant in **N/m**. [2]

Spring constant = ... N/m

(b) The students add another identical spring in parallel with the first one as shown in the diagram.



They find that the spring constant is now doubled. **Add a line on the graph** to show their results. [2]

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